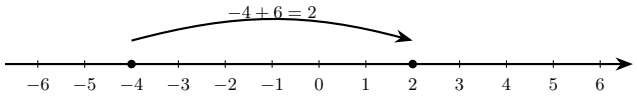
Lesson 1: Review: Integer Operations and Order of Operations

The Grade 9 integer sign rules, then BEDMAS — the building blocks for everything ahead.

1. Adding and Subtracting Integers

Adding a positive moves **right** on a number line; adding a negative (or subtracting a positive) moves **left**. **Same signs:** add and keep the common sign, e.g. $-5 + (-2) = -7$. **Different signs:** subtract the smaller size (distance from zero) from the larger and keep the sign of the number farther from zero, e.g. $-9 + 4 = -5$. Subtracting is adding the opposite: $a - b = a + (-b)$, so $-6 - (-10) = -6 + 10 = 4$.



Example 1

Evaluate each expression.

(a) $-6 + 9$ (b) $-7 - 5$ (c) $8 + (-3)$ (d) $-10 - (-4)$

2. Multiplying and Dividing Integers

Same signs (both positive or both negative) give a **positive** answer: $(4)(3) = 12$ and $(-4)(-3) = 12$. **Different signs** give a **negative** answer: $(-4)(3) = -12$. Division follows the identical rule: $12 \div (-3) = -4$ and $-12 \div (-3) = 4$. Think of each negative sign as flipping the answer once — two flips return you to positive.

Example 2

Evaluate each expression.

(a) $(-4)(6)$ (b) $(-9)(-3)$ (c) $20 \div (-5)$ (d) $(-36) \div (-4)$

3. Order of Operations with Integers

With more than one operation we cannot simply work left to right — follow **BEDMAS**: **B**rackets, **E**xponents, then **D**ivision and **M**ultiplication *together, left to right*, then **A**ddition and **S**ubtraction *together, left to right.* So $-4 + 2 \times (-5) = -4 + (-10) = -14$: multiply first, even though the addition sign is read first.

Example 3

Evaluate each expression, showing each step.

(a) $8 - 2 \times (-5)$ (b) $(-3 + 7) \times (-2) - 1$

4. Exponents with Negative Integers

$(-3)^2$ and -3^2 are **not** the same. Brackets put the negative inside the power: $(-3)^2 = (-3) \times (-3) = 9$. Without brackets the exponent attaches only to the 3, and the negative is applied afterwards: $-3^2 = -(3 \times 3) = -9$. If the negative sign sits outside with no brackets, the answer stays negative.

Example 4

Evaluate each expression.

(a) -4^2 (b) $(-4)^2$ (c) $10 - 3 \times (-2)^2$

5. Putting It All Together

Temperature, elevation, and bank balances all use negative numbers naturally. The last example combines every rule above: signs, exponents, and order of operations.

Example 5

Use everything above. Write the expression first, then evaluate it.
(a) A submarine is at a depth of -45 m. It rises 20 m, then descends 12 m. Write an integer expression for its final depth, then evaluate it.

(b) Evaluate, showing every step in order: $-2 \times (5-8)^2 + (-6) \div 3$

Lesson 2: Review: Operations with Fractions and Decimals

Building, simplifying, and computing with fractions, then converting to and from decimals.

1. Equivalent Fractions and Simplifying

In $\frac{a}{b}$, the **numerator** a counts parts of a whole cut into b equal pieces. Multiplying *both* parts by the same nonzero number builds an **equivalent fraction** ($\frac{2}{3} = \frac{8}{12}$); dividing both by their **greatest common factor** (the biggest number that divides both) reverses it. A fraction is in **lowest terms** once the only common factor left is 1.

Example 1

(a) Fill in the missing numerator to make an equivalent fraction. (b) Simplify to lowest terms.
(a)(i) $\frac{3}{4} = \frac{\quad}{12}$ (a)(ii) $\frac{5}{6} = \frac{\quad}{18}$ (b)(i) $\frac{8}{12}$ (b)(ii) $\frac{10}{15}$ (b)(iii) $\frac{18}{24}$

2. Adding and Subtracting Fractions

Fractions add or subtract only once they share a **common denominator**. Find the LCD (the lowest common denominator, the smallest number both denominators divide into), rewrite each fraction with it, combine the *numerators only* (the denominator stays), then simplify: $\frac{1}{4} + \frac{1}{3} = \frac{3}{12} + \frac{4}{12} = \frac{7}{12}$. With **mixed numbers**, handle the whole-number parts and the fraction parts separately, then recombine.

Example 2

Evaluate. Give each answer in lowest terms.
(a) $\frac{1}{4} + \frac{1}{6}$ (b) $\frac{3}{5} - \frac{1}{4}$ (c) $2\frac{1}{3} + 1\frac{1}{2}$

3. Multiplying and Dividing Fractions

To multiply, go **straight across** — numerators together, denominators together — with no common denominator needed; cancelling shared factors first keeps the numbers small, as in $\frac{3}{4} \times \frac{2}{9} = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. To divide, multiply by the **reciprocal** of the second fraction. **Convert mixed numbers to improper fractions first.**

Example 3

Evaluate. Cancel common factors before multiplying where you can.

(a) $\frac{2}{3} \times \frac{3}{8}$ (b) $\frac{5}{8} \div \frac{3}{4}$ (c) $2\frac{1}{4} \times \frac{2}{3}$

4. Converting Between Fractions and Decimals

Fraction to decimal: divide the numerator by the denominator, so $\frac{3}{4} = 3 \div 4 = 0.75$. **Decimal to fraction:** use the place value of the last digit as the denominator, then simplify: $0.6 = \frac{6}{10} = \frac{3}{5}$. Some quotients end (**terminating**); others, like $\frac{1}{3} = 0.333\ldots$, never do (**repeating**).

Example 4

(a) Convert each fraction to a decimal. (b) Convert each decimal to a fraction in lowest terms.

(a)(i) $\frac{3}{8}$ (a)(ii) $\frac{4}{5}$ (b)(i) 0.45 (b)(ii) 0.85

(c) Convert $\frac{2}{3}$ to a decimal. What do you notice?

5. Order of Operations with Fractions and Decimals

The order never changes: **brackets**, **exponents**, **multiplication and division** left to right, then **addition and subtraction** left to right — so a product hiding inside a sum is done first. When adding or subtracting decimals **line up the decimal points**; when multiplying them, **count the decimal places** in the factors and put that many in the answer. To divide by a decimal, multiply both numbers by 10 until the divisor is a whole number: $1.2 \div 0.3 = 12 \div 3 = 4$.

Example 5

Use the correct order of operations.

(a) Evaluate $\frac{3}{4} - \frac{1}{2} \times \frac{1}{3}$.

(b) A muffin recipe calls for $\frac{2}{3}$ cup of sugar for one full batch. Jordan wants to make *half* a batch. How much sugar should he use?

Lesson 3: Prime Factorization, GCF, and LCM

Breaking numbers into primes, then using those primes to find the GCF and LCM.

1. Prime and Composite Numbers

A **prime number** is a whole number greater than 1 whose only factors are 1 and itself (7 is prime). A **composite number** has more than two factors (12 has $1, 2, 3, 4, 6, 12$). The number 1 is neither — it has only one factor. To test a number, try the primes $2, 3, 5, 7, \ldots$; you can stop once the prime times itself is bigger than the number. Fill in the missing primes below 50:

2, ___, 5, ___, 11, ___, 17, ___, 23, ___, 31, ___, 41, ___, 47

Example 1

Classify each whole number as prime or composite.

(a) 27 (b) 34 (c) 43 (d) 49

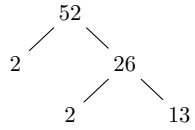
2. Prime Factorization

A **prime factorization** writes a number as a product of primes, usually in exponent form. Two organized methods give the same result: the **division ladder** (divide repeatedly by the smallest prime that fits) and the **factor tree** (split into two factors until every leaf is prime). Both are shown below for 52, giving $52 = 2 \times 2 \times 13 = 2^2 \times 13$.

Division Ladder

2	52
2	26
13	13
	1

Factor Tree



Example 2

Use the method indicated to find each prime factorization. Show every step.

(a) Division ladder: 76 (b) Factor tree: 54

3. Greatest Common Factor (GCF)

The **GCF** is the largest whole number dividing evenly into every number in the group. Factor each number, then take each prime the lists *share*, using its **lowest** power. For $20 = 2^2 \times 5$ and $30 = 2 \times 3 \times 5$, the shared primes are 2 and 5, so GCF $= 2 \times 5 = 10$. For small numbers you can also list the factors: 20: $1, 2, 4, 5, 10, 20$; 30: $1, 2, 3, 5, 6, 10, 15, 30$ $\rightarrow$ GCF $= 10$.

Example 3

Use prime factorization to determine the greatest common factor.

(a) 24 and 42 (b) 45 and 54 (c) 32, 64, and 96

4. Lowest Common Multiple (LCM)

The **LCM** is the smallest whole number that every number in the group divides into evenly. Factor each number, then take *every* prime appearing in *either* list, using its **highest** power. For $14 = 2 \times 7$ and $21 = 3 \times 7$, LCM $= 2 \times 3 \times 7 = 42$. Or list multiples until one matches: 14: $14, 28, 42, \ldots$; 21: $21, 42, \ldots$ $\rightarrow$ LCM $= 42$.

Example 4

Use prime factorization to determine the lowest common multiple.
(a) 10 and 15 (b) 12 and 18 (c) 8, 15, and 20

5. Putting GCF and LCM to Work

GCF problems split things into the *largest* equal groups with nothing left over. **LCM** problems ask when two repeating events line up at the *same time* again.

Examples 5 and 6 — Applications

Solve each problem. State whether you need the GCF or the LCM before you calculate.

5 (GCF). Maya has 36 blue beads and 48 red beads. She wants identical bracelets, using all the beads, with the greatest number of bracelets possible.
(a) How many bracelets can Maya make?

(b) How many blue and how many red beads go on each bracelet?

6 (LCM). Two spotlights sweep a stage. Light A returns to its start every 18 seconds, Light B every 24 seconds. They start together.
(a) After how many seconds are both back at the start together?

(b) How many minutes and seconds is that?

Lesson 4: Classifying Numbers in the Real Number System

Sorting any number into the sets it belongs to, then ordering reals on a number line.

1. Natural Numbers, Whole Numbers, and Integers

The **natural numbers** $N = \{1, 2, 3, \ldots\}$ are the counting numbers; the **whole numbers** $W = \{0, 1, 2, 3, \ldots\}$ add 0; the **integers** $I = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ add the opposites. Each set is tucked completely inside the next, so every natural number is also whole, and also an integer.

Example 1

Given the set $\{-3,\ 0,\ 5,\ -11,\ 14,\ 20\}$, list every member that is:
(a) a **natural** number (b) a **whole** number (c) an **integer**

2. Rational Numbers

A **rational number** Q is any number expressible as a fraction $\frac{a}{b}$ with a, b integers and $b \neq 0$. Every integer qualifies ($9 = \frac{9}{1}$), as does every **terminating** decimal ($0.7 = \frac{7}{10}$) and every **repeating** decimal — so write a number as a decimal and check.

Example 2

Write each as a decimal, say whether it terminates or repeats, and explain why it is rational.
(a) $\frac{3}{8}$ (b) $\frac{4}{9}$ (c) -7 (d) 0.6

3. Irrational Numbers

An **irrational number** $\overline{Q}$ has a decimal that runs on forever without ever settling into a repeating block: $\sqrt{2} = 1.41421356\ldots$, $\pi = 3.14159265\ldots$. A square root (of a whole number) is rational exactly when the radicand is a **perfect square**: $\sqrt{16} = 4$ is rational, but $\sqrt{17}$ is irrational.

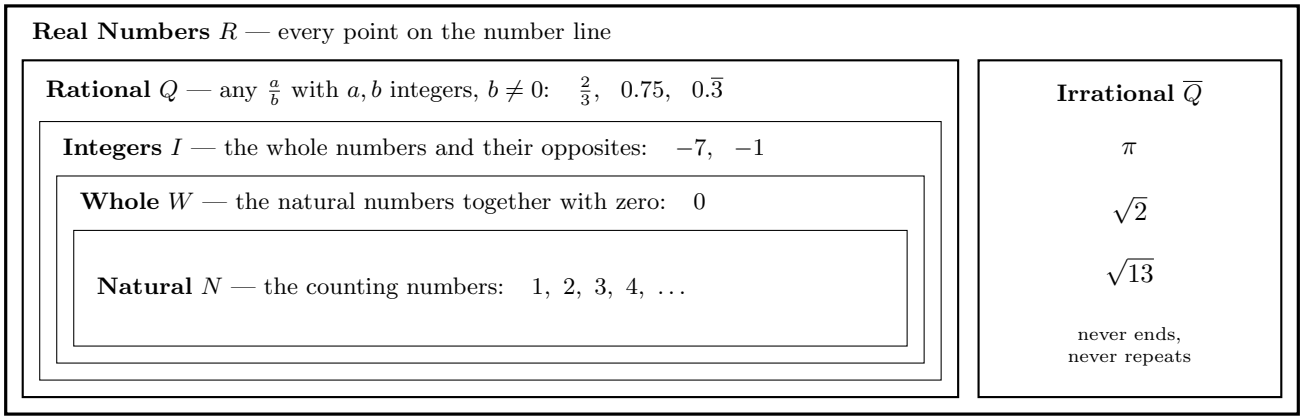
Example 3

Classify each number as rational or irrational, and explain your reasoning.

(a) $\sqrt{2}$ (b) $\sqrt{9}$ (c) π (d) $0.505005000500005\ldots$

4. The Real Number System

Together the rationals and the irrationals fill in *every* point on the number line: this collection is the set of **real numbers** R. The sets nest as $N \subset W \subset I \subset Q \subset R$, with $\overline{Q}$ the only reals outside that chain. A number belongs to *every* set that contains it.



Example 4

Place a checkmark in **every** set each number belongs to.

Number	N	W	I	Q	$\overline{Q}$	R	Number	N	W	I	Q	$\overline{Q}$	R
15							0						
-6							2.4						
$\frac{7}{8}$							$-\sqrt{16}$						
$\sqrt{5}$							$0.\overline{63}$						

5. Ordering and Comparing Real Numbers

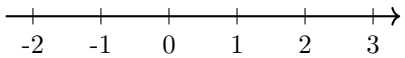
Because the reals fill every point on the line, any two of them can be compared. Estimate an irrational square root from the nearest perfect squares ($\sqrt{2} \approx 1.4$); note $-\sqrt{1} = -1$ is rational, as 1 is a perfect square.

Example 5

Consider $\frac{3}{4}$, $\sqrt{2}$, -1.5, 2, $-\sqrt{1}$, 0.9.

(a) Write the six numbers in order from least to greatest.

(b) Plot each number as a point on the number line below.



Lesson 5: Perfect Squares, Perfect Cubes, and Estimating Roots

Reading roots off the perfect-square list, then trapping the ones that land in between.

1. Perfect Squares and Square Roots

A **perfect square** is a whole number multiplied by itself: $6 \times 6 = 36$, so 36 is a perfect square. The **square root** undoes that, so $\sqrt{36} = 6$. Memorize the list below — it is used constantly for the rest of the course.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

Example 1

Evaluate each square root mentally, using the table above.

(a) $\sqrt{16}$ (b) $\sqrt{81}$ (c) $\sqrt{1}$ (d) $\sqrt{121}$ (e) $\sqrt{49}$ (f) $\sqrt{144}$

2. Perfect Cubes and Cube Roots

A **perfect cube** is a whole number multiplied by itself *three* times: $4 \times 4 \times 4 = 64$. The **cube root** undoes that, so $\sqrt[3]{64} = 4$. A number can be both — $64 = 8^2$ and $64 = 4^3$.

n	1	2	3	4	5	6	7	8	9	10
n^3	1	8	27	64	125	216	343	512	729	1000

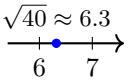
Example 2

Evaluate each cube root mentally, using the table above.

(a) $\sqrt[3]{8}$ (b) $\sqrt[3]{27}$ (c) $\sqrt[3]{125}$ (d) $\sqrt[3]{1}$ (e) $\sqrt[3]{64}$

3. Estimating Square Roots of Non-Perfect Squares

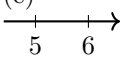
If a number is not a perfect square, **trap** it between the perfect square just below and the one just above; the roots of those give two consecutive whole numbers. Then see which perfect square the number is nearer to. For $\sqrt{40}$: $36 < 40 < 49$, so $6 < \sqrt{40} < 7$. For one decimal, see how far along the gap it is: 40 is 4 of the 13 steps from 36 to 49, about 0.3 of the way, so $\sqrt{40} \approx 6.3$. For cube roots, comparing with the neighbouring cubes is only a rough guide.



Example 3

State the two consecutive whole numbers each root lies between, which one it is closer to, and a one-decimal estimate. In (c), plot your estimate of $\sqrt{27}$ on the number line.

(a) $\sqrt{27}$ (b) $\sqrt{95}$ (c)



4. Using a Calculator to Approximate Roots

On the TI-30XIIS, press 2nd x^2 (the $\sqrt{\ }$ key), the number, then ENTER, and round to the place value asked for: $\sqrt{40} = 6.32455\ldots \approx 6.32$. Always estimate *first* — if the calculator value is far from your estimate, you keyed something in wrong.

Example 4

Evaluate (a) and (b) to the nearest hundredth and compare with your Example 3 estimates.

(a) $\sqrt{27}$ (b) $\sqrt{95}$ (c) Estimate $\sqrt{60}$ by hand, then evaluate it to the nearest tenth. Are the two close?

5. Application: Area and Volume

Roots run the area and volume formulas backwards. A **square** of area A has side length $s = \sqrt{A}$; a **cube** of volume V has edge length $e = \sqrt[3]{V}$.

Example 5

Estimate to the nearest whole number first, then give the calculator value to the nearest hundredth.

(a) A square school garden has an area of 55 m^2. Find its side length.

(b) A cube-shaped storage crate has a volume of 90 cm^3. Find its edge length.

Lesson 6: Simplifying Square Roots: Entire and Mixed Radicals

Pulling perfect squares out from under the root sign, and pushing coefficients back in.

1. Entire Radicals and Mixed Radicals

An **entire radical** has the whole number sitting alone under the root sign: $\sqrt{50}$, $\sqrt{18}$, $\sqrt{7}$. A **mixed radical** has a coefficient outside the root multiplying what remains inside: $5\sqrt{2}$, $3\sqrt{2}$, $4\sqrt{6}$. Every mixed radical can be rewritten as an entire radical; an entire radical can be rewritten as a mixed radical exactly when its radicand has a perfect-square factor other than 1. Check on a calculator that $\sqrt{50}$ and $5\sqrt{2}$ are the same value, then confirm it algebraically:

$$\sqrt{50} = _____ \qquad 5\sqrt{2} = _____ \qquad \sqrt{50} = \sqrt{25 \times ___} = \sqrt{__} \times \sqrt{2} = _____$$

2. Converting an Entire Radical to a Mixed Radical

Split the radicand using $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$, taking the *largest* perfect-square factor and evaluating its root outside. Pulling out a smaller perfect square is not wrong — it just means you must simplify a second time. *Perfect squares:* 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196 (see the Lesson 5 table).

$$\sqrt{48} \;=\; \sqrt{16 \times __} \;=\; \sqrt{__} \times \sqrt{3} \;=\; _____$$

Example 1

Convert each entire radical to a mixed radical in simplest form. Show the factoring step.

(a) $\sqrt{18}$ (b) $\sqrt{20}$ (c) $\sqrt{45}$

3. When the Perfect Square Isn't Obvious

For a large radicand, fall back on the prime factorization from Lesson 3: write the radicand as a product of primes, then move one copy of every *pair* of matching primes outside the root. Any unpaired prime stays inside. For $180 = 2^2 \times 3^2 \times 5$, the pair of 2s gives a 2 outside and the pair of 3s gives a 3 outside, leaving the 5 in: $\sqrt{180} = 2 \times 3 \times \sqrt{5} = 6\sqrt{5}$.

Example 2

Use prime factorization to convert each entire radical to a mixed radical in simplest form.

(a) $\sqrt{126}$ (b) $\sqrt{150}$ (c) $\sqrt{192}$

4. Converting a Mixed Radical to an Entire Radical

Reverse the process: square the coefficient, then multiply it by the radicand already under the root. So $2\sqrt{5} = \sqrt{4} \times \sqrt{5} = \sqrt{4 \times 5} = \sqrt{20}$.

$$2\sqrt{5} \;=\; \sqrt{__} \times \sqrt{5} \;=\; \sqrt{__ \times 5} \;=\; _____$$

Example 3

Convert each mixed radical to an entire radical.
(a) $4\sqrt{3}$ (b) $2\sqrt{13}$ (c) $3\sqrt{7}$ (d) $9\sqrt{2}$

5. Comparing and Ordering Radicals

A mixed radical hides its size behind two numbers, so it is hard to compare at a glance. Convert every radical to entire form, then compare the radicands: the bigger the number under the root, the bigger the root.

Example 4

Without a calculator, arrange the following in order from **least to greatest**: $6\sqrt{2}$, $3\sqrt{5}$, $\sqrt{70}$, $2\sqrt{14}$.

6. Putting It All Together

Two things to watch: a mixed radical is only finished when the number left inside has no perfect-square factor left, and a large radicand may need prime factorization.

Example 5

(a) Sanjay simplified $\sqrt{96}$ this way: $\sqrt{96} = \sqrt{4 \times 24} = 2\sqrt{24}$. Explain why this is *not* fully simplified, then finish simplifying it correctly.

(b) Use prime factorization to simplify $\sqrt{168}$.

Lesson 7: Simplifying Higher-Index Radicals

The same entire–mixed conversion as Lesson 6, now for cube roots and beyond.

1. The Index of a Radical

In $\sqrt[n]{a}$ the small number n is the **index**: it says which root is being taken, and a square root simply hides its index of 2. The Lesson 6 vocabulary carries straight over — a radical is **entire** when a single number sits alone under the root sign ($\sqrt[3]{24}$), and **mixed** when a coefficient multiplies it from outside ($2\sqrt[3]{3}$).

Example 1

State whether each radical is an entire radical or a mixed radical.
(a) $\sqrt[3]{20}$ (b) $4\sqrt[3]{3}$ (c) $\sqrt[3]{9}$ (d) $-5\sqrt[3]{2}$

2. Converting an Entire Cube Radical to a Mixed Radical

Lesson 6 pulled out the largest *perfect square*; with index n you pull out the largest **perfect nth power**, then split with $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$. For example $\sqrt[3]{88} = \sqrt[3]{8 \times 11} = \sqrt[3]{8} \times \sqrt[3]{11} = 2\sqrt[3]{11}$, and 8 was chosen because no larger perfect cube divides 88.

Keep these handy. Perfect cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000. Perfect fourth powers: 1, 16, 81, 256, 625. Perfect fifth powers: 1, 32, 243, 1024.

Example 2

Convert each radical to a mixed radical in simplest form. Show the factoring step.
(a) $\sqrt[3]{40}$ (b) $\sqrt[3]{162}$ (c) $\sqrt[3]{500}$

3. Converting a Mixed Cube Radical to an Entire Radical

To reverse the process, raise the outside coefficient to the power of the index, then multiply it into the radicand: $3\sqrt[3]{7} = \sqrt[3]{3^3} \times \sqrt[3]{7} = \sqrt[3]{27 \times 7} = \sqrt[3]{189}$. Because the index 3 is *odd*, a negative coefficient may be cubed and absorbed as well — unlike a square root, where a negative coefficient must stay outside.

Example 3

Convert each mixed radical to an entire radical. Show the step where you raise the coefficient.
(a) $2\sqrt[3]{9}$ (b) $4\sqrt[3]{2}$ (c) $-2\sqrt[3]{6}$

4. Odd and Even Indices — the Trap

An **odd** index accepts a negative radicand, since an odd power of a negative number stays negative: $\sqrt[3]{-8} = -2$, and $\sqrt[3]{-24} = \sqrt[3]{-8 \times 3} = -2\sqrt[3]{3}$. An **even** index does not: no real number raised to an even power is negative, so $\sqrt{-4}$ and $\sqrt[4]{-16}$ are **not real numbers**. Check the index before you touch a negative radicand.

5. Any Index at All

Nothing about the method changes as n grows — only the list you hunt through: $\sqrt[4]{32} = \sqrt[4]{16 \times 2} = 2\sqrt[4]{2}$ and $\sqrt[5]{96} = \sqrt[5]{32 \times 3} = 2\sqrt[5]{3}$. Going the other way, raise the coefficient to the power of n, not always 3.

Example 4

Convert each radical to the requested form, in simplest form.
(a) $\sqrt[3]{320}$ (mixed) (b) $3\sqrt[3]{13}$ (entire) (c) $\sqrt[4]{243}$ (mixed) (d) $-3\sqrt[3]{10}$ (entire)

Lesson 8: Unit Review: Number Sense

Every Unit 1 rule on one summary page, then mixed practice for the unit test.

1. The Unit 1 Toolkit — Everything from Lessons 1–7

Topic	The rule	Example
Integers & BEDMAS (L1)	Same signs add and keep the sign; different signs subtract and keep the sign of the larger. Subtracting is adding the opposite. Like signs multiply/divide to +, unlike to −. Order: brackets, exponents, ×÷ left to right, +− left to right. Brackets matter: $-n^2 \neq (-n)^2$.	$-5-(-8)=3$ $(-6)(4)=-24$ $-4^2=-16$ but $(-4)^2=16$
Fractions & decimals (L2)	Adding or subtracting needs a common denominator; multiplying does not (cancel common factors first); dividing means multiply by the reciprocal. Fraction to decimal: divide. Terminating decimal to fraction: place value, then reduce.	$\frac{2}{3}+\frac{1}{4}=\frac{11}{12}$ $\frac{3}{4}\div\frac{1}{2}=\frac{3}{4}\times\frac{2}{1}=\frac{3}{2}$ $0.65=\frac{65}{100}=\frac{13}{20}$
Primes, GCF, LCM (L3)	A factor tree breaks a composite number into one unique product of primes. The **GCF** multiplies only the primes in *both* numbers, at the *lowest* shared power; the **LCM** multiplies every prime in *either* number, at the *highest* power present.	$84=2^2\times3\times7$ $\text{GCF}(12,18)=2\times3=6$ $\text{LCM}(12,18)=2^2\times3^2=36$
The number sets (L4)	The sets nest: $N\subset W\subset I\subset Q\subset R$, and $\overline{Q}$ (irrational — a decimal that never ends and never repeats) holds the only reals outside that chain. A root is rational only when the radicand is a perfect power.	7: N, W, I, Q, R $-\sqrt{25}=-5$: I, Q, R $\sqrt{30}$: $\overline{Q}, R$
Radicals (L5–L7)	Entire to mixed: pull out the largest perfect nth-power factor (use prime factorization when it is not obvious). Mixed to entire: raise the coefficient to the power n and multiply it back into the radicand. An odd index absorbs a negative; an even index cannot.	$\sqrt{75}=\sqrt{25}\cdot\sqrt{3}=5\sqrt{3}$ $\sqrt[3]{128}=4\sqrt[3]{2}$ $2\sqrt{11}=\sqrt{4\times11}=\sqrt{44}$

Reference strip — perfect squares and perfect cubes:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
n^3	1	8	27	64	125	216	343	512	729	1000					

Example 1 — Integers and Order of Operations

Evaluate each expression, showing every step in order.

(a) $-15+6\times(-3)$ (b) $(4-9)^2\div(-5)+2$ (c) $-6^2+(-6)^2$

Example 2 — Fractions and Decimals

Evaluate, using the correct order of operations. Cancel common factors first in (b).

(a) $\frac{5}{6}-\frac{1}{4}$ (b) $\frac{3}{10}\times\frac{4}{9}$ (c) $\frac{2}{3}+\frac{1}{2}\times\frac{1}{4}$ (d) Convert 0.35 to a fraction in lowest terms.

Example 3 — Prime Factorization, GCF, and LCM

Use prime factorization for each.

(a) Find the prime factorization of 90. (b) Find the GCF of 40 and 60. (c) Find the LCM of 9 and 15.

Example 4 — Classifying Real Numbers

Place a checkmark in **every** set that each number below belongs to.

Number	N	W	I	Q	$\overline{Q}$	R	**Number**	N	W	I	Q	$\overline{Q}$	R
-10							36						
$\frac{5}{9}$							$-\sqrt{81}$						
$\sqrt{20}$							$0.\overline{27}$						

Example 5 — Perfect Powers and Estimating Roots

(a) Evaluate $\sqrt{100}$ and $\sqrt[3]{216}$ mentally.

(b) Estimate $\sqrt{70}$: state the two consecutive whole numbers it lies between, state which one it is closer to, and give a one-decimal estimate.

Examples 6 and 7 — Simplifying Radicals

Write each radical in the form asked for, in simplest form.

6. (a) Convert $\sqrt{80}$ to a mixed radical. (b) Use prime factorization to convert $\sqrt{147}$ to a mixed radical. (c) Convert $6\sqrt{5}$ to an entire radical.

7. (a) Convert $\sqrt[3]{135}$ to a mixed radical. (b) Convert $4\sqrt[3]{3}$ to an entire radical. (c) Convert $\sqrt[4]{162}$ to a mixed radical.

Challenges 1 and 2 — Mixed Skills

1. Simplify $\sqrt{242}$ to a mixed radical. Then state whether the simplified value is rational or irrational, and explain how you know.

2. A rectangular field measuring 75 m by 90 m is going to be tiled with the largest possible identical square sections, with no leftover strip along either side.

(a) Use the GCF of 75 and 90 to find the side length of the largest possible square section.

(b) A second square section has an area of 75 m^2. Is its side length rational or irrational? Estimate it to the nearest tenth of a metre.